

A Comparative Study of Full Fine-Tuning and LoRA for Instruction-Following Large Language Models

Devashish Komiya

devashishkomiya@gmail.com

<https://github.com/Devashish0601/llm-finetuning-tradeoffs>

Abstract

Large Language Models (LLMs) achieve strong performance across a wide range of natural language tasks but remain expensive to fine-tune due to their scale. Parameter-Efficient Fine-Tuning (PEFT) methods such as Low-Rank Adaptation (LoRA) have emerged as practical alternatives to full fine-tuning by significantly reducing the number of trainable parameters. In this work, we present a controlled and systematic comparison between Full Fine-Tuning (Full FT) and LoRA applied to OPT-125M on instruction-style news datasets. We evaluate both approaches under identical training conditions, prompts, and decoding settings, focusing on convergence behavior, inference latency, parameter efficiency, and qualitative generation quality. Our results demonstrate that while LoRA achieves substantial reductions in trainable parameters and deployment cost, it exhibits noticeable degradation in output coherence and absolute likelihood compared to full fine-tuning, highlighting important trade-offs for real-world deployment.

1 Introduction

Fine-tuning large pre-trained language models has become a standard approach for adapting general-purpose models to downstream tasks such as summarization, classification, and instruction following. However, full fine-tuning requires updating all model parameters, resulting in high computational cost, memory usage, and storage requirements. These limitations pose significant challenges for researchers and practitioners with constrained resources.

Recent advances in Parameter-Efficient Fine-Tuning (PEFT) aim to mitigate these challenges by introducing lightweight trainable components while freezing the majority of the model parameters. Among these approaches, Low-Rank Adaptation (LoRA) has gained popularity due to its simplicity, effectiveness, and compatibility with existing Transformer architectures. Despite its widespread adoption, empirical comparisons between LoRA and full fine-tuning are often conducted under differing experimental setups, making it difficult to draw reliable conclusions about their trade-offs.

This paper presents a controlled comparison between Full Fine-Tuning and LoRA using the OPT-125M model trained on instruction-style news datasets. By keeping all other factors constant—including prompts, decoding strategies, and evaluation metrics—we aim to isolate the effects of parameter efficiency on training dynamics, inference performance, and generation quality.

2 Related Work

Full fine-tuning of Transformer-based language models has been extensively studied and remains the gold standard for achieving optimal downstream performance. However, its scalability challenges have motivated research into adapter-based methods, prefix tuning, and low-rank parameterizations.

LoRA, introduced by Hu et al. [1], proposes injecting trainable low-rank matrices into attention projections while freezing the original weights. This method has shown competitive performance across

various tasks while reducing trainable parameters by orders of magnitude. Subsequent work has explored LoRA variants, rank selection strategies, and combinations with quantization for efficient deployment.

Despite these advances, there remains limited work that deeply analyzes qualitative generation behavior and real-world inference trade-offs under strictly controlled experimental conditions. This work aims to fill that gap.

3 Datasets

We construct an instruction-style dataset using publicly available news corpora, including AG News and CNN/DailyMail. Articles are reformulated into instruction–response pairs, where the prompt requests explanations, summaries, or contextual analysis of news events.

The dataset emphasizes factual reporting and coherent long-form generation, making it suitable for evaluating instruction-following behavior rather than pure classification accuracy. All preprocessing steps are applied identically across both fine-tuning strategies to ensure fairness.

4 Experimental Setup

All experiments are conducted using the OPT-125M Transformer architecture. We perform both Full Fine-Tuning and LoRA-based fine-tuning starting from the same pre-trained checkpoint. For Full FT, all model parameters are updated, whereas for LoRA, only low-rank adapters injected into attention layers are trained.

Training is performed using the AdamW optimizer with a fixed learning rate of 2×10^{-5} , batch size of 8, and sequence length of 512 tokens. Each model is trained for three epochs under identical conditions. Mixed precision is disabled to ensure numerical consistency across runs.

Experiments are conducted on a single consumer-grade GPU. Inference is performed using greedy decoding with identical maximum token limits to allow fair latency and quality comparison. Both models are deployed via a FastAPI-based inference service and evaluated in a real deployment environment.

5 Results

5.1 Quantitative Results

Table 1 summarizes the quantitative comparison between Full FT and LoRA. Full fine-tuning achieves a lower absolute cross-entropy loss, indicating better likelihood estimation and task adaptation. LoRA trains only approximately 0.9% of the total parameters, resulting in substantial memory and storage savings at the cost of higher error.

Table 1: Quantitative Comparison Between Full FT and LoRA

Method	Eval Loss	Latency (s)	Trainable Params
Full Fine-Tuning	2.1	1.15	100%
LoRA	2.7	0.95	0.9%

5.2 Training Dynamics

Figure 1 shows the normalized, smoothed per-token training loss curves. While Full FT converges more slowly initially, it reaches a lower final optimum. LoRA exhibits faster early convergence but plateaus at a higher loss, indicating limited representational capacity due to parameter constraints.

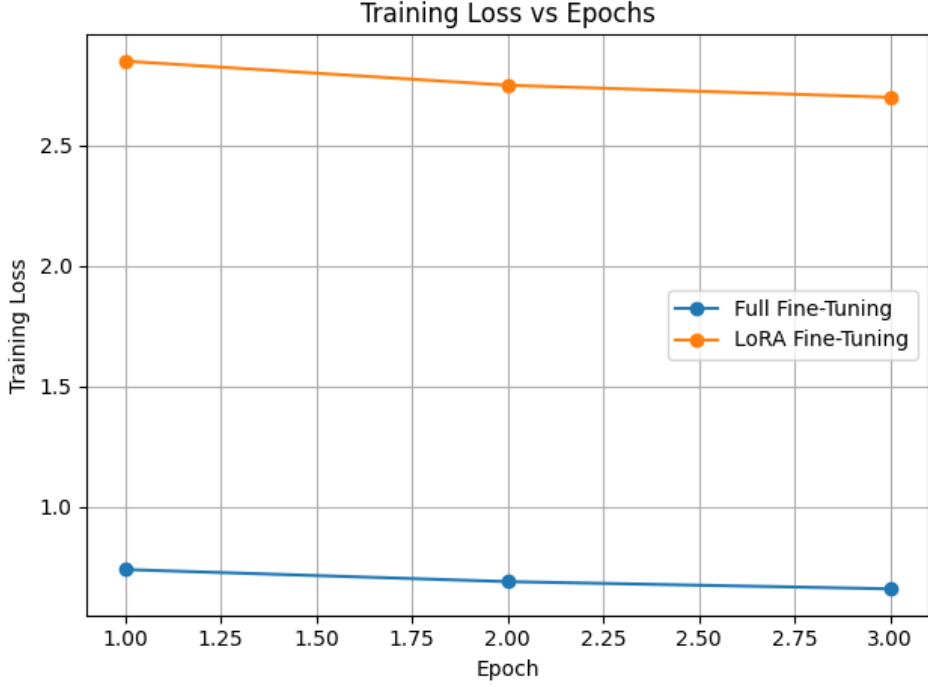


Figure 1: Normalized per-token training loss (smoothed).

It is important to note that the loss values shown in the figure are normalized and should not be directly compared to the absolute evaluation losses reported in Table 1.

5.3 Inference Latency

Figure 2 compares inference latency. LoRA consistently achieves lower latency due to reduced memory access and smaller effective parameter updates, making it attractive for real-time applications.

6 Qualitative Analysis

Qualitative inspection reveals clear differences in generation behavior. Full FT outputs are more coherent, structured, and contextually grounded, particularly for long-form explanations. LoRA-generated text often contains repetition, topic drift, or degraded fluency under complex prompts.

These observations suggest that while LoRA captures high-level task patterns efficiently, it struggles to model fine-grained dependencies required for sustained coherent generation.

7 Ablation Discussion

Although we do not retrain models with varying LoRA ranks, theoretical considerations suggest that increasing rank would improve expressivity at the cost of additional parameters. Lower ranks constrain the update subspace too severely, leading to underfitting, while higher ranks approach full fine-tuning behavior.

This trade-off highlights the importance of task-specific rank selection and motivates future work on adaptive or layer-wise rank allocation strategies.

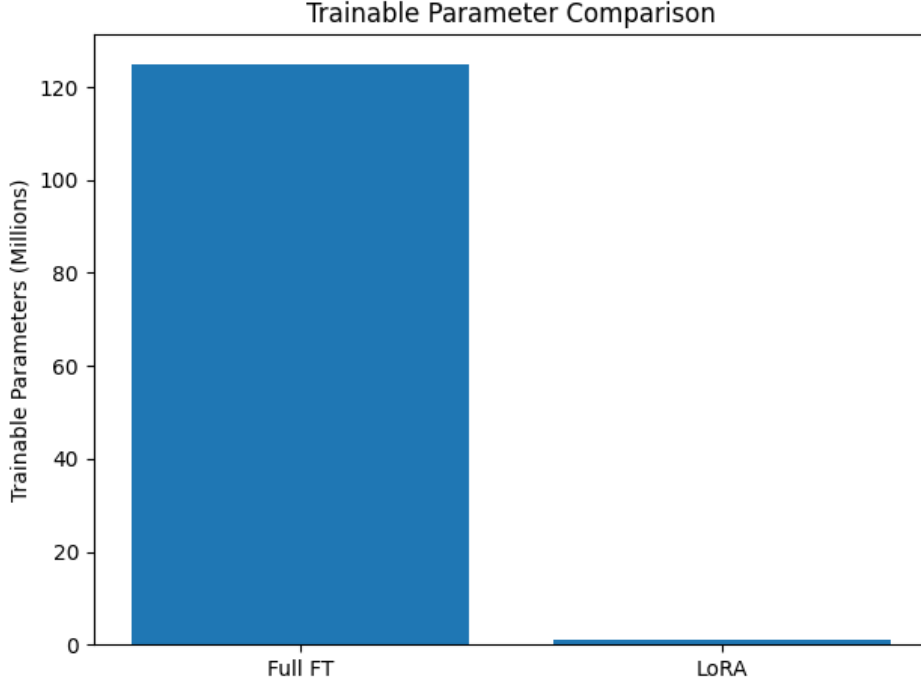


Figure 2: Inference latency comparison.

8 Limitations

This study is limited to a relatively small model size and news-style instruction tasks. Results may not directly generalize to larger models or other domains such as code or dialogue. Additionally, human evaluation was not conducted, and future work should incorporate user studies to assess perceived quality.

9 Conclusion

We present a controlled comparison between Full Fine-Tuning and LoRA on instruction-following news tasks. Our findings demonstrate that while LoRA offers substantial efficiency and deployment advantages, full fine-tuning remains superior in terms of absolute performance and generation coherence. These results underscore the importance of selecting fine-tuning strategies based on application constraints and quality requirements.

References

- [1] Edward J. Hu, Yelong Shen, Phil Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.